

WASP-39b

R-1185

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_230327 · created 2026-07-31T04:59:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460030.8703 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1494 +/- 0.0082
Transit depth (Rp/Rs)^2	2.23 +/- 0.25 [%]
Orbital Inclination (inc)	87.76 +/- 0.67
Airmass coefficient 1 (a1)	0.9778 +/- 0.0092
Airmass coefficient 2 (a2)	0.018 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#2 - [62, 180]
Optimal Aperture	4.38
Optimal Annulus	10.36
Transit Duration (day)	0.1196 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1494 $\pm$ 0.0082 vs 0.1457 (deviation 0.45 σ)
Duration fit vs published	0.1196 d vs 0.12 d (deviation 0.06 σ)
Mid-time O-C	-4.7 min
Depth SNR	18.2
Residual scatter	0.94 %

Light curve

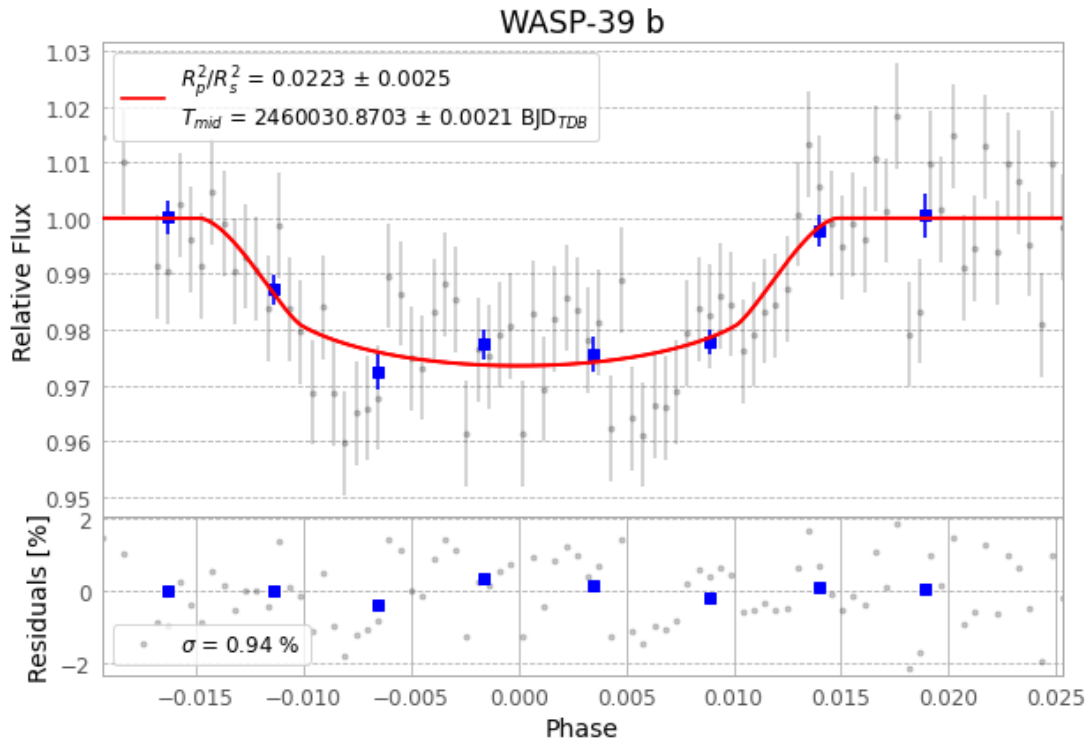


Figure 1 — FinalLightCurve_WASP-39 b_2023-03-26.png (SHA-256 5415477ee7704070...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 108], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 575528e2af0b94e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), WASP-39230327065116.FITS → WASP-39230327111215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-230327.log (72fcab6594e8...), FinalLightCurve_WASP-39 b_2023-03-26.png (5415477ee770...), FinalLightCurve_WASP-39 b_2023-03-26.csv (b531a9a8632e...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:59Z.